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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 9

Title: Exploratory data analysis on business datasets

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Experiment No.: 9

Exploratory data analysis on business datasets

Problem Statement

You are a Data Analyst at a consulting firm. A client has provided a large business dataset and wants to understand key trends and patterns. Perform exploratory data analysis (EDA) to uncover meaningful business insights and identify opportunities for improvement.

Introduction

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset using statistical methods, data visualization, and analytical techniques. It helps identify patterns, relationships, trends, and unusual observations before performing advanced analysis.

In this experiment, a synthetic dataset containing **500 student records** was analyzed using Power BI. The dataset contains information about students' study habits, attendance, coding practice, assignment completion, learning platforms, AI tool usage, mock tests, and performance scores.

An interactive one-page dashboard was developed to present the important findings in a simple and professional format. The dashboard uses KPI cards, charts, and slicers to help users explore student learning patterns and their relationship with academic performance.

Objectives

The main objectives of this experiment are:

1. To understand the structure of a student learning dataset.
2. To analyze student learning behavior.
3. To calculate important performance indicators.
4. To examine relationships between learning activities and performance.
5. To compare student performance across academic years.
6. To analyze the usage of different learning platforms.
7. To identify different student performance segments.
8. To create meaningful data visualizations.
9. To develop an interactive Power BI dashboard.
10. To derive useful insights from the analyzed data.

Methodology / Steps Followed

The project was implemented using **Microsoft Power BI**. The major steps followed were:

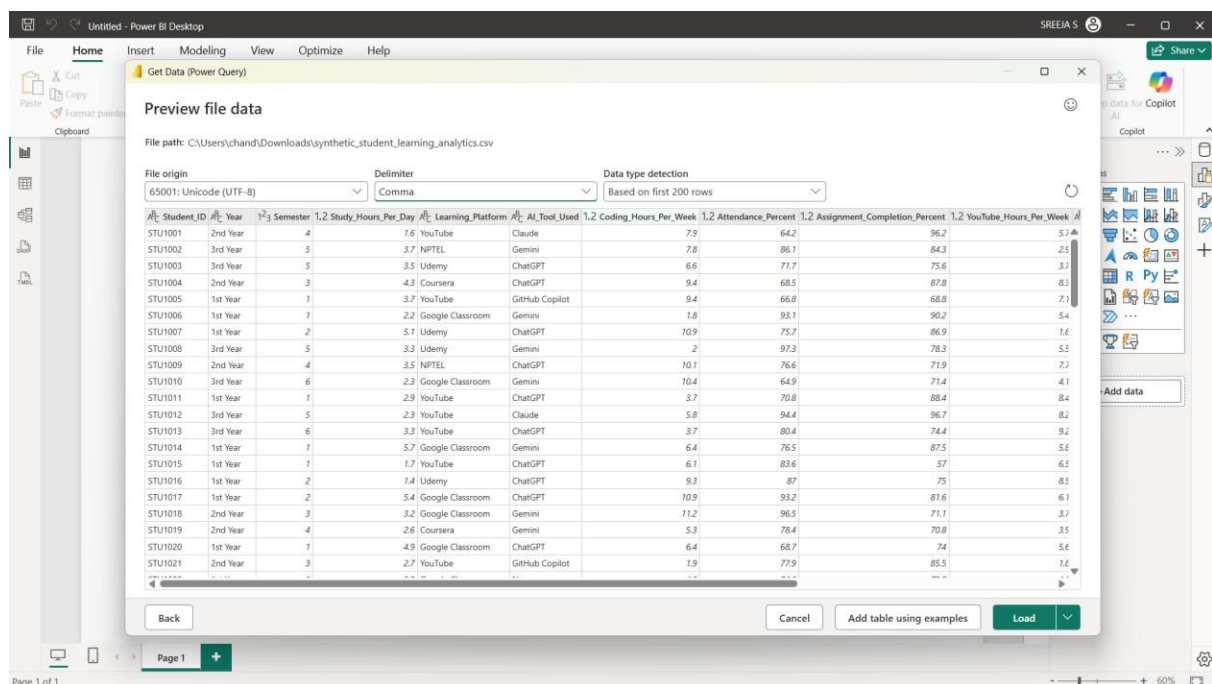
1. Dataset Import
2. Data Cleaning and Transformation
3. Data Type Validation
4. DAX Measure Creation
5. KPI Creation
6. Dashboard Development
7. Interactive Filtering
8. Data Visualization
9. Insight Generation

Step 1: Dataset Import

The synthetic student dataset was imported into **Power BI Desktop** using the **Text/CSV** connector.

The process involved:

1. Opening Power BI Desktop.
2. Selecting **Get Data**.
3. Selecting **Text/CSV**.
4. Selecting the student dataset.
5. Previewing the data.
6. Loading the dataset into Power BI.



Step 2: Data Cleaning and Transformation

The dataset was checked to make sure that the data was suitable for analysis.

The following were checked:

- Column names
- Missing values
- Data types
- Numerical fields
- Categorical fields
- Student performance values

The data was prepared before creating the dashboard.

Step 3: Data Type Validation

The important columns were assigned appropriate data types.

Column	Data Type
Student_ID	Text
Year	Text
Semester	Whole Number
Study_Hours_Per_Day	Decimal Number
Learning_Platform	Text
AI_Tool_Used	Text
Coding_Hours_Per_Week	Decimal Number
Attendance_Percent	Decimal Number
Assignment_Completion_Percent	Decimal Number
YouTube_Hours_Per_Week	Decimal Number
Group_Study	Text
Mock_Tests_Per_Month	Whole Number
Performance_Score	Decimal Number
Student_Segment	Text

Step 4: DAX Measure Creation

DAX measures were created to calculate important student performance indicators.

Total Students

Total Students =

DISTINCTCOUNT(Student_Data[Student_ID])

Average Performance

Average Performance =

AVERAGE(Student_Data[Performance_Score])

Average Attendance

Average Attendance =

AVERAGE(Student_Data[Attendance_Percent])

Average Study Hours

Average Study Hours =

AVERAGE(Student_Data[Study_Hours_Per_Day])

Average Coding Hours

Average Coding Hours =

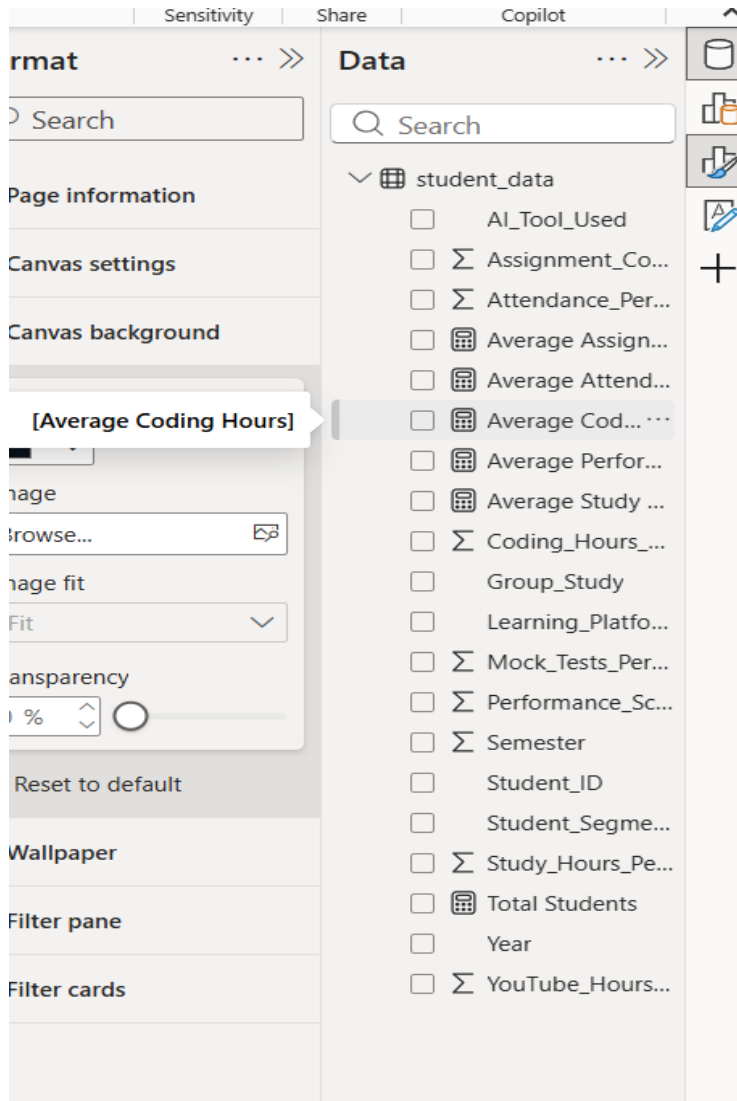
AVERAGE(Student_Data[Coding_Hours_Per_Week])

Step 5: KPI Creation

Five KPI cards were created:

- **Total Students**
- **Average Performance**
- **Average Attendance**
- **Average Study Hours**
- **Average Coding Hours**

These KPIs provide a quick overview of student learning and performance.



Step 6: Dashboard Development

A professional **one-page Student Learning Analytics Dashboard** was developed using Power BI.

The dashboard contains:

- KPI cards
- Interactive slicers
- Performance analysis
- Study analysis
- Attendance analysis
- Learning platform analysis
- Student segment analysis

Step 7: Interactive Filters

The following slicers were added:

- Year
- Semester
- Learning Platform
- Student Segment

These filters allow users to analyze specific groups of students.

Step 8: Data Visualization

Six important charts were created.

8.1 Average Performance by Year

Chart: Clustered Column Chart

- Axis: Year
- Value: Average Performance

This chart compares student performance across different academic years.

8.2 Study Hours vs Performance

Chart: Scatter Chart

- X-axis: Study Hours Per Day
- Y-axis: Performance Score
- Details: Student ID
- Legend: Student Segment

This chart shows the relationship between study time and performance.

8.3 Attendance vs Performance

Chart: Scatter Chart

- X-axis: Attendance Percentage
- Y-axis: Performance Score
- Details: Student ID

This helps analyze the relationship between attendance and performance.

8.4 Student Performance Segments

Chart: Donut Chart

- Legend: Student Segment
- Values: Total Students

The students are divided into:

- High Performer
- Average Performer
- Needs Improvement

8.5 Students by Learning Platform

Chart: Bar Chart

- Axis: Learning Platform
- Value: Total Students

This shows the most commonly used learning platforms.

8.6 Coding Practice and Performance by Year

Chart: Combo Chart

- Axis: Year
- Column: Average Coding Hours
- Line: Average Performance

This compares coding practice and performance across academic years.

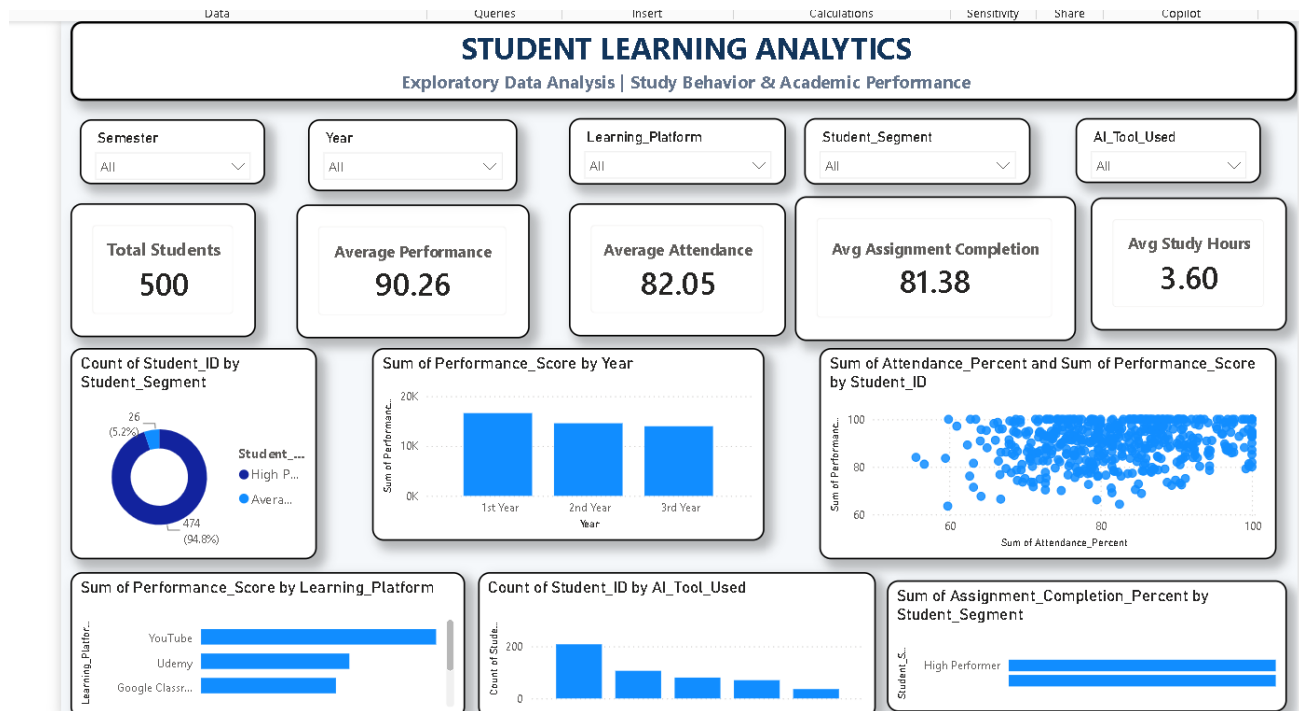
Step 9: Dashboard Design

A professional **dark theme** was used for the dashboard.

The design includes:

- Dark background
- Professional header
- KPI cards
- Blue and teal accents
- Clear chart titles
- Interactive slicers
- Consistent layout

The one-page dashboard provides a simple overview of student learning behavior and performance.



Business Questions Answered

The dashboard helps answer:

1. How many students are included in the dataset?
2. What is the average student performance?
3. What is the average attendance?
4. How many hours do students study?
5. Which year has better performance?
6. Is study time related to performance?
7. Is attendance related to performance?
8. Which learning platform is most popular?
9. What percentage of students are high performers?
10. How does coding practice vary across years?

Key Insights

The dashboard can be used to identify:

- Differences in performance between academic years.
- Relationship between study hours and performance.
- Relationship between attendance and performance.
- Most commonly used learning platforms.
- Distribution of high, average, and low-performing students.
- Differences in coding practice between academic years.

Conclusion

The **Student Learning Analytics Dashboard** demonstrates how Power BI can be used to explore and understand student data.

The dataset was imported, prepared, and analyzed using DAX measures and visualizations. KPI cards were used to show important student statistics, while charts were used to analyze performance, study hours, attendance, learning platforms, and coding practice.

Interactive slicers allow users to filter the dashboard and analyze different student groups. The final dashboard provides a simple and professional view of student learning patterns and performance.

Learning Outcomes

Through this experiment, the following skills were developed:

- Power BI data import
- Data cleaning
- Data type validation
- DAX measure creation
- KPI creation
- Data visualization
- EDA
- Student performance analysis
- Interactive slicers
- Dashboard design
- Insight generation
- Data-driven decision-making

GitHub Repository Link:

https://github.com/Balajivallepu/Data_Analytics- for Bussiness

GitHub Project Folder Link:

https://github.com/Balajivallepu/Data_Analytics- for Bussiness/tree/main/EXPERIMENT9